

Metadata

Source

Sheldon Axler, *Measure, Integration & Real Analysis*.

Area

Measure Theory

Type

Personal study notes and solved exercises.

Status

Personal exposition containing original explanations, intuitive interpretations, and independently written solutions.

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Purpose

To develop rigorous intuition for outer measure and the transition from Riemann integration to abstract measure theory while studying graduate-level analysis.

Abstract

This project presents a systematic mathematical study of the Riemann Integral and its limitations in measuring a set, which eventually leads to the Outer measure and its properties. The Outer measure creates the building block before setting the axiomatic, abstract and more advanced tools for the Measure Theory. Thus, this project is crucial for developing the true intuitions prior to the rigorous measure theory, a cornerstone in modern mathematics. It contains key theorems, definitions, and proofs, reflecting my own understanding and originality. Although most of the mathematical sections in this project are truly inspired from the graduate level textbook *Measure, Integration & Real Analysis* by *Sheldon Axler*, this project only accumulates my own concretized versions of this reference. This project ends with the key exercises from the same book of Axler, with the detailed solution distilled with emphasis and intuition. This exercise section developed through independent study and problem solving. The transition from Riemann integration to outer measure is explored as a foundational step toward abstract measure theory

1 Riemann Integral

The celebrated Riemann integral, used in finding the area under the curves, carries conceptual sequences that bridge the gaps to truly understand the traditional integration. I will start with the most important thing essential to grasp the Riemann integral. [Partition] A partition is used in the desired closed interval $[a, b]$, where $a, b \in \mathbb{R}$ and $a < b$, for making finer sub-divisions of $[a, b]$. If we consider a finite list $x_0, x_1, \dots, x_n$, inside the primary interval, then it gives a finite ordered set to be called partition:

$$a = x_0 < x_1 < x_2 < \dots < x_n = b$$

This partition can truly now capture the whole interval $[a, b]$, and can be imagined this closed interval as finite union of these closed sub-intervals $[x_{j-1}, x_j]$, as

$$[a, b] = \bigcup_{j=1}^n [x_{j-1}, x_j]$$

Simply: finer the partition, better will be the accuracy. For capturing any function in $[a, b]$ with the highest details, intuitively, finest partition is ideal. As moving forward, the profoundness of partition will be revealed. Throughout next, the notation P will be used referring finite sequence for partition. I imagine it as a knife, where we are looking for a sharpest knife and the most skillful hand to chop inside the interval as fine as possible. Hence, sharper the knife in a skillful hand, allowing us to capture more detail of the function on $[a, b]$.

Definition 1.1 (Lower Riemann Sum). A lower Riemann sum $L(f, P, [a, b])$ is used to capture the bounded function $f : [a, b] \rightarrow \mathbb{R}$ with each sub-intervals of the partition as possible lowest height rectangles. It gives the total area of the rectangles, covering the interval $[a, b]$, where the height of each rectangle is the infimum of the function f and the width as $[x_{j-1}, x_j]$. Lower Riemann sum is defined as

$$L(f, P, [a, b]) = \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f$$

Definition 1.2 (Upper Riemann Sum). An upper Riemann sum $U(f, P, [a, b])$ keeps the similar concept of $L(f, P, [a, b])$, just with the modification of the heights of the rectangles. Here, the heights of the rectangles are the supremum of the function f in each sub-intervals $[x_{j-1}, x_j]$. Hence, the definition will be

$$U(f, P, [a, b]) = \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f$$

[Lower and Upper Riemann Sum.] The true objective of lower and upper Riemann sums is to evaluate the area under the function in a maximum efficient way. Lower Riemann sum represents area under the curve from just the below of the function. Whereas, the upper Riemann sum represents the area from just above the function. The rectangles in lower Riemann sum occupy little less area than the true area and the reverse happens in the case of upper Riemann sum. Hence

$$L(f, P, [a, b]) \leq U(f, P, [a, b])$$

and the equality is trivial for any constant function, as $\inf f = \sup f$. The purpose for these two sums is to make this inequality to equality. The only thing that could do that is partition or P . If one makes the partition finer, lower Riemann sum increases and upper Riemann sum decreases and continuing the inclusion of partition P , keeping the interval same, can eventually make the two sums equal. This is the true beauty of Riemann sums for defining the Riemann integral.

Definition 1.3 (Lower Riemann Integral). As the previous concept pushes that our objective is to increase the lower Riemann sum for closely capturing the total area of the bounded function in the interval $[a, b]$, lower Riemann integral $L(f, [a, b])$ is defined as

$$L(f, [a, b]) = \sup_P L(f, P, [a, b])$$

The suffix P below supremum confirms that this can only be achieved by controlling the supremum, which actually, to tighten the partition. Here, the supremum is taken over all partitions P of $[a, b]$.

Definition 1.4 (Upper Riemann Integral). Upper Riemann integral represents the infimum of the Upper Riemann sum. Hence

$$U(f, [a, b]) = \inf_P U(f, P, [a, b])$$

The upper Riemann sum is decreased with the tightening of the partition, which is desired. Here the infimum is taken over all partitions P of $[a, b]$.

It is easy to visually verify the following theorem regarding lower and upper Riemann integral.

Theorem 1.5. A bounded function $f : [a, b] \rightarrow \mathbb{R}$. Then

$$L(f, [a, b]) \leq U(f, [a, b])$$

This leads to decide when a bounded function can be called as Riemann integrable. Both lower and upper Riemann integral are intuitively very close to the true area. But one equality imposition will make it most accurate and end this confusion.

Definition 1.6 (Riemann Integrable). A bounded function on a closed bounded interval is called *Riemann Integrable* if its lower Riemann integral equals its upper Riemann integral.

$$L(f, [a, b]) = U(f, [a, b])$$

This is the popular integration of a function f bounded in a closed interval $[a, b]$

$$\int_a^b f = L(f, [a, b]) = U(f, [a, b])$$

[Functions Not Riemann Integrable] This section will highlight the fact that this mathematically precise definition of Riemann integral is not applicable to all functions. The most appreciated example is the Dirichlet function.

Dirichlet Function: The Dirichlet function $f(x)$ is

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

If $[a, b] \in \mathbb{R}$ & $a < b$, then $\exists t \in [a, b]$ such that $t \in \mathbb{Q}$ and, $\exists s \in [a, b]$ such that $s \notin \mathbb{Q}$. Take an interval $[0, 1]$ and the partition $P = x_0, x_1, \dots, x_N$ with each sub-interval $[x_{j-1}, x_j]$. Then

$$\sup_{[x_{j-1}, x_j]} f = 1$$

and

$$\inf_{[x_{j-1}, x_j]} f = 0$$

This must be true as f is Dirichlet function, which means in every sub-interval $[x_{j-1}, x_j]$, there is a rational number and an irrational number (Real analysis theorem). Thus $f = 1$ and $f = 0$ belong to every sub-interval. Hence, lower Riemann sum is zero and upper Riemann sum is 1 for each sub-interval. Thus lower Riemann integral and upper Riemann integral are not equal. Hence, the Dirichlet function is not Riemann integrable.

1.1 Failure of Riemann Integral

1. Riemann integral fails for functions with many discontinuities such as the Dirichlet function in the previous example.
2. Riemann integral fails for unbounded functions and strictly applicable for bounded continuous functions.
3. Doesn't work well with pointwise limits like the Dirichlet function.

These insist a requirement for more precise theory of integration and the Riemann integral, although very strong, is not ideal. This opens the door for *Measure Theory*, which is an abstract and most accurate theory of integration for any types of functions. The intuitive clarity of measure theory can only be developed from a true understanding of the *Outer Measure*. This is what I am starting with in the next section.

2 Outer Measure

Notation. Throughout this section, the outer measure of a set $A \subset \mathbb{R}$ will be denoted by (A) .

Definition 2.1. The outer measure of a set $A \subset \mathbb{R}$ is defined by

$$(A) = \inf \left\{ \sum_{k=1}^{\infty} l(I_k) : I_1, I_2, \dots \text{ are open intervals such that } A \subset \bigcup_{k=1}^{\infty} I_k \right\}$$

where $l(I_k)$ is the length of each open interval that covers from the outside of the set A .

[Infinite Union of Open Intervals as Covers] Basically, the key objective of defining the outer measure of a set is to capture the set from outside using open intervals that contain the set, as tightly as possible. Stress on the fact of infinite union of those intervals. So, the question comes: Why infinite unions? This is for generalization purpose. If the set behaves very bad (like Cantor-type), where finite open covers will always give gap, one needs infinitely large number of covers. It is better to imagine the open cover as ϵ -neighborhood. Caution: All intervals must be from the outside of the set. Hence, open intervals fit beautifully to define outer measure. Entering inside the set is **STRICTLY NOT ALLOWED!** The another beauty of this is the choice of these open intervals is as arbitrary as possible as long as they are outside the set A . Thus

- These open intervals can overlap

- Can actually do anything being in outside, containing the set A
- They have all freedom and flexibility from outside the set. This precisely help to capture very bad set

If one imagining a very oscillating subset of $\mathbb{R}$, he can choose a single point in that ugly set and create a cover that is precisely close to that point and also containing the whole set. And continuing for all those risky points similarly to cover them. Eventually the best cover using infimum operation will arise which will be the outer measure of that set. It is easy now to visualize all those open covers will badly overlap among each other, and that is absolutely fine. Imagine packing a gift. We can only pack from the outside of the gift. The goal is to pack as efficiently as possible. This theoretically means that if we try packing in several ways, the best one will be the infimum of total lengths of covering by the packing. This will capture any arbitrary shape by the packing by minimizing the gaps from boundary of the gift.

$$\text{cover from outside} \implies \inf \text{ over all coverings}$$

This infimum of all packing is exactly what is outer measure of a set. All mathematical machinery will be eventually discussed in this section.

3 Some Nice Properties of Outer Measure

Theorem 3.1. Every countable subset of $\mathbb{R}$ has outer measure 0.

The main idea here (and most outer measure proofs/problems) is to accurately define the open cover of each countable set. A countable set means it contains distinct elements inside the set. Hence, each element can be covered individually without influencing other. A perfect countable set can be $A = \{a_1, a_2, \dots, a_k\} \subset \mathbb{R}$, where each element are singleton $\{a_k\}$. An ϵ -neighborhood of open interval around each $\{a_k\}$ is absolutely feasible. The mature question should come here: what should be the most ideal way to define this interval containing $\{a_k\}$? This is

$$(a_k - \frac{\epsilon}{2^k}, a_k + \frac{\epsilon}{2^k})$$

Why $\frac{\epsilon}{2^k}$ and not only ϵ ?

This is the key understanding. A singleton in a countable set gives trouble when capturing using the open covers. The reason is, a singleton is exactly a point and one needs infinitely close interval to capture in a most close way. This $\frac{\epsilon}{2^k}$ helps to take infinite sum of the intervals, as with each sum over k , reduces the length of the interval and comes extremely close to $\{a_k\}$. The sequence tells that the infinite sum is like $1 + \sum_{n=1}^{\infty} \frac{1}{n}$, $\forall n \in \mathbb{N}$. Which is just 1. Hence the infinite sum of all these intervals

$$\sum_{k=1}^{\infty} l(I_k) = 2\epsilon$$

So from the definition of outer measure

$$(A) \leq 2\epsilon$$

Now, ϵ is an arbitrary positive number. Thus, $(A) = 0$. This completes the proof.

Remark 3.2. The 2^k in the denominator of the neighborhood makes only sense here because it is around a singleton. So in the length of interval, $\{a_k\}$ vanishes. That is

$$l(I_k) = (a_k + \frac{\epsilon}{2^k}) - (a_k - \frac{\epsilon}{2^k}) = \frac{2\epsilon}{2^k}$$

and remains only ϵ term which will not trouble at all with infinite sum. But this would terribly fail for any other subset of $\mathbb{R}$ (eg. (a, b)) as the interval would be $(b - a) + \frac{2\epsilon}{2^k}$. And this infinite sum would only give ∞ as the infinite sum of any finite interval would give only ∞ .

[Outer Measure of Countable Set] The previous theorem that the outer measure of any countable set of $\mathbb{R}$ is 0, carries significant weight. Intuitively, one cannot make a nonzero cover from the collection of open intervals to capture point like elements. Points are truly zero length structure and outer measure zero strengthens this very fact. This actually says, any finite set of real numbers(points), will always give outer measure 0. More specifically, any set of distinct numbers will always give zero total outer measure. Any open interval for covering a singleton is squeezing to zero. This is also one of the drawbacks of outer measure as will be explored later when I will explain Lebesgue measure. One can generate another theorem from the above argument for a very important subset of real numbers.

Theorem 3.3. The Outer measure of $\mathbb{Q}$ is always 0

Thus, one can also say that $(\mathbb{R} \cap \mathbb{Q}) = 0$ (as used in one of the exercises). The next theorem is extremely important to establish a relationship between two sets with the outer measure, when we know that one set is subset of another.

Theorem 3.4 (Monotone Theorem). If $A, B \subset \mathbb{R}$ and most importantly $A \subset B$, then

$$(A) \leq (B)$$

As $A \subset B$, every open cover of B , quite intuitively, must also contain A . Hence

$$(A) \leq \sum_{k=1}^{\infty} l(I_k)$$

where right side of the inequality represents the cover of set B . Taking infimum over all sequences of open intervals whose union contains B , one can easily get $(A) \leq (B)$.

Definition 3.5 (Translation Preserves Outer Measure). For any scalar $t \in \mathbb{R}$, and $A \subset \mathbb{R}$, translating the set A by t is basically

$$t + A = \{t + a : a \in A\}$$

Now, the key thing in the basis of outer measure, is that length of an open or closed interval (theorem comes later), cannot be affected by translating the set along the $\mathbb{R}$. If one is not modifying the set, the interval will give exactly the same infimum over all lengths of the cover, i.e. the outer measure will remain exactly the same, by translating with some scalar amount along the $\mathbb{R}$.

This gives another property of the outer measure, which can be mentioned as a theorem.

Theorem 3.6. Outer measure is translation invariant.

If $t \in \mathbb{R}$ and $A \subset \mathbb{R}$, then

$$(t + A) = (A)$$

Although, the intuition is not hard enough for translation invariance of outer measure, the proof needs a trick! Consider a set $A \subset \mathbb{R}$ then

$$(t + A) \leq \sum_{k=1}^{\infty} l(t + I_k) = \sum_{k=1}^{\infty} l(I_k) = (A)$$

The main trick is to prove the opposite inequality. Writing the set A as

$$A = -t + (t + A)$$

taking outer measure both sides

$$(A) = (-t + (t + A))$$

Now using above inequality

$$(-t + (t + A)) \leq (t + A)$$

This is obvious as $(-t) = 0$ as $-t$ is just a singleton. Hence

$$(A) \leq (A)$$

Thus, it proves the equality

$$(t + A) = (A)$$

Example 3.7. Take intervals $(2, 7)$ and $(6, 10)$.

$$l((2, 7) \cup (6, 10)) = l(2, 10) = 8$$

and the sum of individual interval lengths

$$l(2, 7) + l(6, 10) = 5 + 4 = 9$$

The key reason for this strict inequality is that $(2, 7)$ and $(6, 10)$ are not pair wise disjoint. This will give clarity with sub-additivity theorem.

Theorem 3.8 (Countable Subadditivity of Outer Measure).

$$\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} (A_k)$$

The trick for this proof is

$$\sum_{j=1}^{\infty} l(I_{j,k}) \leq \frac{\epsilon}{2^k} + (A_k)$$

It is intuitively very clear, especially for non-disjoint sets, the open cover and the infimum of the total length of the intervals, should always be less than the infimum length of the open intervals for individual set separately considering. This is countable subadditivity of outer measure. It is applied to sequences of subsets.

4 Outer Measure of Closed Bounded Interval

Now, we have reached to the interval which is the only concerned interval for Riemann integration. This section will precisely show why the outer measure for a closed interval $[a, b]$ where $a < b$, is $b - a$. The proof of this will again claim the beauty of open ϵ -neighborhood to determine outer measure. The key understanding for this proof is open cover. Open cover has been mentioning throughout this article for deliberately defining outer measure. But a proper definition will be nice.

Definition 4.1 (Open Cover). If $A \subset \mathbb{R}$ then a collection $\mathcal{C}$ of open subsets of $\mathbb{R}$ is called an open cover of A if A is contained in the union of all sets in $\mathcal{C}$. But this open cover is not unique, and there can be several ways we can use open cover to contain a set.

The first inequality to prove that

$$([a, b]) \leq b - a$$

If one takes sequence of open intervals $(a - \epsilon, b + \epsilon), \emptyset, \emptyset, \dots$, whose union contains $[a, b]$, then

$$([a, b]) \leq b - a + 2\epsilon$$

Now this holds true for all $\epsilon > 0$

$$([a, b]) \leq b - a$$

The main and challenging inequality to prove is the opposite one. This needs the celebrated *Heine-Borel* theorem.

4.0.1 Celebrated Heine-Borel Theorem

Theorem 4.2 (Heine-Borel Real Analysis Version). Say, $A \subset \mathbb{R}$ and A is compact if and only if A is bounded and closed.

$\Leftarrow$ If A is bounded, any sequence $(a_n) \in A$ must have a converging subsequence (Bolzano-Weierstrass Theorem). Hence accumulation point or limit point exists. This guarantees convergence. Hence is compact.

$\Rightarrow$ A is compact. Hence a sequence $(a_n) \in A$ and $\lim(a_n) = a \in \mathbb{R}$. If we can prove that $a \in A$, it will prove that A is closed. Now, as A is compact—it has an accumulation or limit point. Say, $a \in A$. As a convergence sequence can have only one limit point

$$\lim(a_n) = a \in A$$

Thus A is closed. Next we need to show that A is bounded. Let's use contraposition. If A is not bounded, then a sequence $(a_n) \in A$. So, from the definition of unboundedness

$$|a_n| > n, \forall n \in \mathbb{N}$$

But as we already found that (a_n) cannot have any accumulation or limit point value, it contradicts our finding of unique limit point. Hence A is bounded. Both way proof done.

Theorem 4.3 (Heine-Borel Measure-theoretic or Topological version). Every open cover of a closed and bounded subset of $\mathbb{R}$ has a finite subcover.

Remark 4.4. This above measure-theoretic version of Heine-Borel theorem is exactly saying the same thing of real analysis version of it. Thus, it exactly defines *Compactness*. It basically says $[a, b]$, which is compact, then any open cover of $[a, b]$ that contains $[a, b]$, must have a finite subcover to capture it. Outer measure for any closed interval $[a, b]$ is defined with the inequality of open cover i.e. sum of finitely many open intervals or lengths.

$$([a, b]) \leq \text{Sum of finitely many open intervals or lengths}$$

This is exactly *Heine-Borel* theorem or compactness, borrowed from real analysis. Thus, Heine-Borel theorem says that if $[a, b]$ is compact, then finite subcover exists and hence $\implies$

$$[a, b] \subset \bigcup_{k=1}^N I_k$$

Some key significances of Heine-Borel theorem are

- Every continuous function on compact set must be uniformly continuous.
- Continuous functions on compact set attains maximum and minimum values.
- Convergence and limits.
- A set in $\mathbb{R}$ is sequentially compact iff it is closed and bounded (compact).
- In metric spaces, Heine-Borel property is not generally true.

[Classic Proof of Heine-Borel Theorem] **Bisection Method:** Take a closed and bounded interval $[a, b]$. We need to show it is compact. Say, no finite subcover. Let's start with a relatively large open interval I_1 . Then repeatedly halve the interval. We can always take some point c which will always be in the intersection of all finite intervals. But c is in some open set of covers and, that open set covers a neighborhood of c . So it has a finite subcover. Contradicts our assumption.

Theorem 4.5 (Outer measure of A closed Interval). Using Heine-Borel theorem and previous inequality proof: if $a, b \in \mathbb{R}$, with $a < b$, then

$$([a, b]) = b - a$$

5 Outer Measure is Not Additive

This is the crucial stage we reached, where it reflects now the weakness of the outer measure. This theorem reveals the imperfection of the outer measure for determining the absolute measurability of a set. Nonadditivity, without any doubt, is the strongest weakness of the outer measure for a set.

Theorem 5.1. There exist disjoint subsets A and B of $\mathbb{R}$ such that

$$(A \cup B) \neq (A) + (B)$$

The outer measure of the union is not at all related to the outer measures of the individual sets. The most intuitive saying is that outer measures do not care about the structure of the set. It only tries to fit as efficiently as possible from the outside of the set. It is a type of cheapest cover. Hence, it doesn't respect the local behavior of the set. And during union of the sets, we are more interested in the local behavior during the cover. It is very likely with some visualization that local behavior of the set after union of the sets, maybe completely different and unrelated with local behavior of the individual parts. Recalling the main definition of the outer measure, it gives infinite freedom and flexibility for the open covers to capture the set as long as it contains the whole set. So, the tightest possible fit can be achieved from outside. But union of them mostly always give a situation where a different outside packing structure (or outer measure) arises. Hence equality and additivity, fails in outer measure.

Example 5.2. $A = \{0\}$ and $B = \{1\}$. Both are singleton. Hence

$$(A) = (B) = 0$$

Thus

$$(A) + (B) = 0$$

Now take the union

$$A \cup B = \{0, 1\}$$

For the outer measure of $A \cup B$, take most precise open cover $(-\frac{\epsilon}{2}, 1 + \frac{\epsilon}{2})$. hence the length of the interval is $I = 1 + \epsilon$. Now with the infimum of this interval I gives

$$(A \cup B) = \inf I = \inf(1 + \epsilon) = 1$$

So we found that

$$(A \cup B) \neq (A) + (B)$$

Additivity of the outer measure fails after taking union.

6 Solutions of Exercise 2A (Axler) With My Insights

Prove that if A and B are subsets of $\mathbb{R}$ and $(B) = 0$, then $(A \cup B) = (A)$.

Proof. Using countable subadditivity

$$(A \cup B) \leq (A) + (B)$$

As $(B) = 0$

$$(A \cup B) \leq (A) \tag{1}$$

Now think, if there is a way I can reverse this inequality. Because the prove demands equality. using the monotone theorem, it is clear that

$$A \subseteq A \cup B$$

Hence

$$(A) \leq (A \cup B) \tag{2}$$

Thus, using (1) and (2):

$$(A \cup B) = (A)$$

□

Suppose $A \subset \mathbb{R}$ and $t \in \mathbb{R}$. Let $tA = \{ta : a \in A\}$. Prove that $(tA) = |t|(A)$

Proof. Case I: Trivial case

$A = \emptyset, t = 0$. Then $(tA) = (\emptyset) = 0 = |t|(A)$

$$A = \emptyset, t \neq 0$$

Then

$$(tA) = (\emptyset) = 0 = |t|(A)$$

$$A \neq \emptyset, t = 0$$

Then

$$(tA) = (\{0\}) = 0 = |t|(A)$$

Case II:

$$t \neq 0, A \neq \emptyset$$

Taking countable collection of open intervals $\{I_n\}$ and $\epsilon > 0$. Then

$$A \subseteq \bigcup_{n=1}^{\infty} I_n.$$

Now, the length of the interval I_n satisfies

$$l(I_n) < (A) + \epsilon$$

This strict inequality is always possible to create as $\epsilon > 0$. Now, length of each scaled interval is

$$l(tI_n) = |t|l(I_n)$$

Thus,

$$l(tI_n) = |t|l(I_n) < |t|(A) + \epsilon$$

Taking the infimum on the length of interval

$$(tA) \leq |t|(A) + |t|\epsilon$$

Now, $\epsilon > 0$ is arbitrary, and hence

$$(tA) \leq |t|(A)$$

Now, for the reverse inequality, we need to prove

$$(tA) \geq |t|(A)$$

Or, rearranging

$$(A) \leq \frac{1}{|t|}(tA)$$

This rearranging is the key step in this proof. Now, we can definitely replace A with tA and t with $\frac{1}{t}$ in the previously obtained inequality. this gives

$$\left(\frac{1}{t}(tA)\right) \leq \left|\frac{1}{t}\right|(tA)$$

Thus

$$(tA) \geq |t|(A)$$

So, as both sides inequalities are established, we found

$$(tA) = |t|(A)$$

□

This equality gives an important theorem regarding the scaled outer measure of a subset of $\mathbb{R}$. So, a scaled outer measure of a subset is just the multiplication of the absolute magnitude of the scalar with the outer measure of the subset.

And regarding the proof for the nontrivial case, the reverse inequality implements a trick to rewrite the inequality, replacing the subset A with tA and, scalar t with $\frac{1}{t}$. This is preserving the structure of the first inequality and gives the reverse inequality, which forces the equality. Prove that if $A, B \subset \mathbb{R}$ and $(A) < \infty$, then $(B \setminus A) \geq (B) - (A)$

Proof. Rearranging the given statement to verify, will immediately lead to a smooth track.

$$(B \setminus A) + (A) \geq (B)$$

Now, a hint of subadditivity has automatically come! The job is if we can prove this or not. For that it is needed to check the sets only. From the basic set theory

$$B \setminus A \cup A = B \cup A$$

$\Rightarrow$

$$(B \setminus A) + (A) \geq B \cup A$$

It is obvious

$$B \subseteq B \cup A$$

Hence from the monotonicity theorem of outer measure:

$$(B) \leq (B \cup A)$$

Using these two inequalities, it is certain

$$(B \setminus A) + (A) \geq (B)$$

Thus

$$(B \setminus A) \geq (B) - (A)$$

Things to note: This proven inequality holds true only if $(A) < \infty$. Because if $(A) = \infty \Rightarrow (B \cup A) = \infty$. But

$$\infty \not\geq \infty$$

Hence, we could only claim the equality not the strict inequality if $(A) = \infty$. Thus the full inequality can only be established if the subset is finite. □

The outer measure of a subset excluded from another subset of $\mathbb{R}$ greater than or equal to outer measure of the excluded subset subtracted from the outer measure of the another subset, gives an intuitive image. Say, when A and B are disjoint. Then $(B \setminus A) = (B) > (B) - (A)$ if A has a finite nonzero outer measure. The equality is guaranteed from this example, when $(A) = 0$. Better visualization exercise is when we can take A and B with some partial overlapping. Now, it is clear that individual outer measure from the infimum, the infimum of the length of unions for subset A will be little more as it will be counted twice. This gives the understanding of the strict inequality. The simple observation that $B \subseteq B \cup A$ is essential to use the monotonicity and eventually to establish the main inequality using subadditivity. The finiteness of the outer measure of subset A is must for proving the inequality. But infinite outer measure of subset A cannot say anything about the strict inequality. Prove that if $a, b \in \mathbb{R}$ and $a < b$ then

$$((a, b)) = ([a, b]) = ((a, b]) = b - a$$

Proof. The key finding is the proper open cover for these intervals. The best possible open cover, with an ϵ -neighborhood, for all of these sets is

$$(a - \epsilon, b + \epsilon)$$

Then the length of the interval is

$$I_k = b - a + 2\epsilon$$

Taking infimum

$$(A) = b - a$$

where A represents each of the above subsets. □

This is key result for the outer measure of differently types of intervals, including open, closed, half-open or half-closed. From infimum of open covers to include the intervals where the limit points are same, always give the difference of them as the outer measure. Suppose a, b, c, d are real numbers with $a < b$ and $c < d$. Prove that

$$((a, b) \cup (c, d)) = (b - a) + (d - c)$$

if and only if $(a, b) \cap (c, d) = \emptyset$

Proof. $\implies ((a, b) \cup (c, d)) = (b - a) + (d - c)$. Using the subadditivity

$$((a, b) \cup (c, d)) \leq ((a, b)) + ((c, d))$$

Now, $((a, b)) = b - a$ and $((c, d)) = d - c$. Thus

$$((a, b) \cup (c, d)) \leq (b - a) + (d - c)$$

If $(a, b) \cap (c, d) \neq \emptyset$, then

$$((a, b) \cup (c, d)) < (b - a) + (d - c)$$

This strict inequality as sum of each outer measure would always be more than the union, as evident from the infimum of interval length concept. But it is given that $((a, b) \cup (c, d)) = (b - a) + (d - c)$. Hence $(a, b) \cap (c, d) = \emptyset$.

$$\Longleftarrow (a, b) \cap (c, d) = \emptyset$$

Taking an open cover of (a, b) as $(a - \epsilon, b + \epsilon)$, the length of the open cover is

$$l(I_{(a,b)}) = b - a + 2\epsilon$$

Taking infimum gives $b - a$. Similarly

$$\inf l(I_{(c,d)}) = d - c$$

Hence

$$((a, b)) + ((c, d)) = (b - a) + (d - c)$$

The left side of above equality is only because they are disjoint and thus union is basically sum of the outer measures. □

Explicitly, if two intervals intersect, they together lie in a single interval, whose measure strictly less than the sum of the individual outer measure. The equality only comes if they are disjoint, as the measure of the union is exactly the same of taking infimum of open interval lengths for each and sum them. This is the main understanding from this proof. Prove that if $A \subset \mathbb{R}$ and $t > 0$, then $(A) = (A \cap (-t, t)) + (A \cap (\mathbb{R} \setminus (-t, t)))$

Proof. From my initial observation, what I sense: if I can show that $A = (A \cap (-t, t)) \cup (A \cap (\mathbb{R} \setminus (-t, t)))$ and both $(A \cap (-t, t))$ and $(A \cap (\mathbb{R} \setminus (-t, t)))$ are disjoint, then from subadditivity of outer measures, the job is done. This is exactly lead me to the full proof. A set is measurable if it splits every set additively via outer measure. This, what I proved using pure set-theoretic language, can actually be rigorously shown using the Lebesgue measure theorems, especially the Caratheodory's extension theorem. $\square$

Prove that $([0, 1] \setminus \mathbb{Q}) = 1$

Proof. The key observation here is that the known measure is $([0, 1]) = 1$. So, if this can be written in the form of $[0, 1] \setminus \mathbb{Q}$. Which definitely possible.

$$[0, 1] = ([0, 1] \setminus \mathbb{Q}) \cup ([0, 1] \cap \mathbb{Q})$$

Applying outer measure

$$([0, 1]) = (([0, 1] \setminus \mathbb{Q}) \cup ([0, 1] \cap \mathbb{Q}))$$

In the first problem, it was proved that

$$(A \cup B) = (A)$$

if $(B) = 0$. Hence,

$$(([0, 1] \setminus \mathbb{Q}) \cup ([0, 1] \cap \mathbb{Q})) = (([0, 1] \setminus \mathbb{Q}) + ([0, 1] \cap \mathbb{Q}))$$

As $([0, 1] \setminus \mathbb{Q}) \cap ([0, 1] \cap \mathbb{Q}) = \emptyset$ and

$$([0, 1] \cap \mathbb{Q}) = (\mathbb{Q}) = 0$$

Thus,

$$(([0, 1] \setminus \mathbb{Q}) \cup ([0, 1] \cap \mathbb{Q})) = (([0, 1] \setminus \mathbb{Q}) + (\mathbb{Q})) = ([0, 1]) = 1 - 0 = 1$$

$\square$

$\mathbb{Q}$ is an countable set and outer measure is zero. It is a dense set with outer measure zero. Whereas, $\mathbb{R}$ is a dense set with outer measure nonzero.